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SUMMARY

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■ Research Objective: The Need for Data Platform Activation to Support Enterprise AI

- AI diffusion accelerates enterprise innovation, elevating firms' data capabilities as a core determinant of AI model and service development

  - Data has evolved into a strategic corporate asset and a core object of market transactions

- SMEs and ventures face three structural constraints: (ⅰ) limited access to high-quality data, (ⅱ) insufficient regulatory compliance capacity, and (ⅲ) shortages of specialized talent

  - Such disparities risk cumulative competitive erosion and market exit for SMEs in the AI-driven digital economy

- Data platforms should function as institutional mechanisms that assure quality and standards while reducing search and compliance costs

■ Research Methodology

- This paper analyzes the current state and limitations of enterprise data utilization and derives policy measures for activating data platforms

- It first examines the status and characteristics of domestic data platforms*, analyzing limitation from the perspective of data utilization

* AI-Hub (NIA), Data Voucher (K-DATA), Healthcare Big Data Platform (HIRA), Financial Data Exchange (FSI), National Transportation Data Open Market (ex), and Pitagraph (private)

- It further analyzes the data strategies of leading jurisdictions(U.S., Europe, China, Japan)to derive implications for the domestic ecosystem

■ Limitations of Domestic Data Platforms

- (Supply-demand mismatch) Supply-centric construction misaligns with demand, propagating bottlenecks into the utilization stage

  - Leading reason for non-use remains "no data worth using" (53.2%); pain points have shifted to utilization-stage issues

- (Demand-side capacity gaps) Shortages of in-house expertise and technical infrastructure for data collection, analysis, and processing impede firms from initiating utilization

  - Parallel support for capabilities and workforce is required

- (Quality and utilization infrastructure) Project-based construction limits quality management; absent standards inflate verification and preprocessing costs

  - PETs hold potential for sensitive-data use, yet adoption is constrained by underdeveloped legal frameworks and validation criteria

- (Pseudonymized data and secure zones) Rigid linkage procedures and prolonged reviews delay data access for startups and other users

  - Provider risk aversion, over-pseudonymization, and on-site-only analysis erode data utility

- (Private-sector-led linkage and integration) Fragmented sectoral standards undermine demand-driven linkage and constrain private-led ecosystem formation

- Public platform linkage should be complemented by private platform cultivation to promote demand-based distribution and integration

■ Implications from International Policy Landscape

- (United States - market) Lowers entry barriers via negative regulation and opt-out regimes, cultivating an organic market led by private data brokers and hyperscalers
- (Europe - regulation) Establishes institutional foundations for data sovereignty and trust, fostering a federated ecosystem in which standards-based common data spaces enable safe cross-industry data flows
- (China - state) Designates data as a core factor of production, with strong state governance complemented by private-sector technological capabilities in a state-directed ecosystem
- (Japan - public-private cooperation) Employs flexible norms such as government certifications and guidelines to induce private-sector autonomy, pursuing an open strategy oriented toward global interoperability

Policy Direction 1. Capability Dimension

- (Combination·integration utilization) As linkage and fusion drive utilization value, continued discovery of high-demand datasets should be coupled with cross-domain integration of existing assets
- (Strengthening utilization capacity) Enhance enterprise data literacy for precise use-case definition, and datafy field-level tacit expertise to translate AI use into productivity gains
- (AI-Ready data) Refine and scale AI-Ready standards for formats, metadata, and quality frameworks to ensure immediate AI usability

Policy Direction 2. Infrastructure Dimension

- (Trust and fairness) Deploy fair-competition mechanisms to induce open market structures, reinforcing standard contracts and dispute resolution as effective transactional safeguards
- (Quality and standardization) Platforms should embed linkage-ready acquisition and quality management upfront, with national standards for terminology, schemas, and transmission protocols
- (Diffusion of PETs) To expand sensitive-data use as training inputs, support private-sector PETs development and prioritize their application and piloting, with technical assistance to lower enterprise adoption barriers

Policy Direction 3. Legal and Institutional Dimension

- (Privacy protection and use) Institutional refinements are required for effective pseudonymized-data use, including clearer interpretive guidance, streamlined ex ante review, and statutory coherence
- (Institutional instruments) Enhance ex ante legal certainty via no-action letters and sectoral Q&A frameworks, and expand regulatory sandboxes to foster predictable utilization
- (Easing review burden) Standardize pseudonymization and review criteria, and expand one-stop support and delegation to specialized agencies to shorten processing timelines

- (Reassessing pseudonymized public data) Reassess the efficacy and risks of releasing pseudonymized public data, with consideration of tiered operations calibrated to purpose and risk level

Policy Direction 4. Ecosystem Dimension

- (Unifying the supply pipeline) A data catalog search and source-information infrastructure is in place (One-Window), but wider catalog diffusion is needed for diverse private-sector demands such as cross-industry data combination

  - Core attributes (custodianship, format, refresh cycle, terms of use) are codified as standard metadata; mandatory inter-platform linkage and further advancement should follow

  - Priority datasets in high demand for AI should be selected and systematically developed, managed, and supplied via sector-specific platforms

- (Staged ecosystem development) Build platform-centric ecosystems in phased stages tailored to sectoral characteristics, reinforcing institutions, infrastructure, and standards

  - Diffuse a federated model of anchor firms defining problems and hyperscalers providing platforms, with public seed support in public-interest domains